A SPHERICAL MIXTURE MODEL APPROACH FOR 360 VIDEO VIRTUAL CINEMATOGRAPHY

Chenglei Wu[⋆] *Zhi Wang*[‡] *Lifeng Sun*[⋆†∗]

[⋆] Department of Computer Science and Technology, Tsinghua University
[†]BNRist, Dept. of Computer Science and Technology, Tsinghua University
[∗]Key Laboratory of Pervasive Computing (Tsinghua University), Ministry of Education, China
[‡] Tsinghua Shenzhen International Graduate School, Tsinghua University

ABSTRACT

360 video virtual cinematography attempts to direct a virtual camera and capture the most salient regions of 360 videos. In this paper, we propose a data-drive solution to achieve high-quality and diversified 360 cinematography based on crowd-sourced viewing histories. Specifically, we try to address two problems: 1) how to locate the semantically important regions of interest (RoI) from raw data, 2) how to generate virtual camera paths that follow chronological narratives. We first design a dynamic spherical mixture model based algorithm to locate variable number of RoIs on each video frame. We then model the camera transition and chronological orders with a Bayesian network and conditional probabilities. With the above two designs, we can generate "optimal" cinematography paths based on a dynamic programming algorithm. By modeling the RoIs as spherical mixture model, we are also able to provide diversified cinematography results. We demonstrate the effectiveness of our algorithm through extensive experiments.

Index Terms— 360 Video; Virtual Cinematography

1. INTRODUCTION

360 virtual cinematography attempts to direct a virtual camera and capture the most important and interesting field of views (FoV) of a 360 video, following a chronological order. Such technology empowers many appealing applications, for example, 360 video auto playback, which automatically choose the viewport of a 360 video for user and relieves users from head movements or mouse clicks.

The state-of-the-art 360 virtual cinematography solutions [1, 2] often approach this problem from video content perspective, e.g., using deep learning techniques. However, as mentioned in [1], these methods can only be applied to a very limited domain of videos, i.e., single RoI 360 videos. When applied to other domains of videos, such as multi-RoI videos (e.g., Fig. 1) or no-RoI videos (sightseeing videos, e.g., Fig. 2), these methods often either fail to locate the correct RoIs and/or move the virtual camera abruptly.

Fig. 1: Mutli-RoI Video

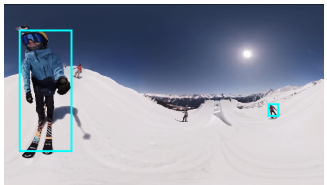

Fig. 2: Sightseeing Video

In this paper, we propose to achieve high-quality and diversified 360 cinematography in a crowdsourced manner. Specifically, we try to answer two questions: 1) As users' attention distributions on 360 videos are highly dynamic and distributed, how can we correctly estimate the number and locations of the RoIs on each video frame? 2) how can we discover and model the human attention transition patterns resulted from the progresses of the events, thus generating virtual camera paths that follow the chronological narrative?

1.1. Contribution

To address the first problem, we design a dynamic spherical attention clustering algorithm: We infer the number and locations of multiple RoIs on 360 videos by grouping similar attention regions or viewports in a time window, thus each cluster corresponding to a unique RoI. Due to the spherical nature of 360 videos, we model the location of a RoI on a 360 video frame with von Mises-Fisher distribution (vMF)[3] on a 3D unit sphere. We utilize vMF mixture model (vMFMM) to dynamically locate the multiple RoIs on 360 video frames.

To address the second problem, i.e. capturing the chronological orders revealed in the viewing traces, we model users' attention transitions with Bayesian network. The chances of transition between RoIs are considered as the causal relations and modeled as conditional probabilities. The joint conditional probabilities of RoI transitions along a virtual camera path can then be calculated and used to evaluate if this path follows the the logic and chronological order.

With the above two methods, we are able to choose the

viewport on each video frame. We further apply a Kalman smoother to the generate trace to smooth camera movements. By modeling RoIs with vMFMM probability model, viewport on each video frame can be sampled from the probability distribution, thus providing diversified user experiences. To evaluate the effectiveness of our framework, we compare our algorithms to convention solutions, such as spherical DBSCAN, and present a set of demo videos to show the excellent virtual cinematography results. To our best knowledge, this is the first crowdsourced method that achieves practical and high-quality 360 virtual cinematography.

1.2. Related Work

Hu et al. [1] propose to utilize Faster R-CNN object detection network to locate candidate objects on a video frame, and train a object selector and a path regressor to move a virtual camera between different objects. Su et al. [2] extend the AutoCam [4] method to allow more general and dynamic camera controls. Jayaraman et al. [5] adopt a deep reinforcement learning (DRL) based method to train an agent that can explore unseen 360 environments. The problem with these work is that they fail to take into account the logic order and narrative of a 360 video.

2. BACKGROUND AND FRAMEWORK

We use the dataset contributed by Wu et al.[6] for experiments, which contains the head movement traces of 48 people viewing 18 videos, including sport, nature scene, show etc.

2.1. Motivation

Current content-based methods can be applied to specific domain of videos, i.e., single RoI 360 videos. In [1], the authors demonstrate that the content-based method often abruptly moves the virtual camera on equally salient multi-RoI 360 videos[1]. And on no-RoI sightseeing 360 videos, in which the background view is more important than the foreground video shooter, content-based methods only focus on the computational salient area, i.e., the video shooter. What's worse, content-based methods fail to understand the narratives of video contents, e.g., the story of a 360 Sci-Fi video, thus unable to generate corresponding camera movements. The drawbacks make these methods impractical and unfeasible for real-world deployment. We thus propose crowdsourced virtual cinematography method to cope with the problems.

2.2. Spherical Mixture Model based RoI Location

2.2.1. vMF Basics

To achieve accurate clustering of spherical data, we model the location of RoI on 360 video as vMF distribution. The vMF distribution is commonly used to model directional data on sphere. It is parameterized by a mean direction μ and a concentration scalar $\tau > 0$. The probability of a direction vector $n \in R^3$ is defined as:

$$f(n|\mu, \tau) = \frac{\tau e^{\tau \mu^T n}}{2\pi(e^{\tau} - e^{-\tau})} \tag{1}$$

In our approach, we model a RoI's position on a 360 video as the center of vMF distribution, and the range of this RoI is represented by the concentration.

However there are cases that a 360 video frame has multiple RoIs or the viewing directions are highly distributed, which can't be modeled with a single vMF distribution. We thereby adopt the vMF mixture model (vMFMM), which fits the data with K vMF distributions. The probability of n in vMFMM is defined as:

$$f(n|\Theta) = \sum_{h=1}^{K} a_h f(n|\mu_h, \tau_h) \tag{2}$$

where $\Theta = \{a_h, \mu_h, \tau_h | 1 \leq h \leq K\}$ and $\sum_{h=1}^{K} a_h = 1$.

2.2.2. Dynamic vMFMM Clustering Algorithm

In our approach, we group N users' viewport in a time window into different clusters, i.e., vMF distributions, and the centers of the distributions are considered as the locations of the RoIs on 360 videos in that time window. Users' viewports are represented by the forward vectors, i.e., viewing directions. The duration of time window is T and the data sample frequency is f. In this paper, we use a duration $T = 0.25s$ and sample frequency $F = 16Hz$. Therefore the number of viewports in each time window is to $N \times F \times T$.

The parameters of vMFMM can be estimated with expectation-maximization (EM) algorithm. The inference process requires the prior knowledge of the number of mixtures K, which, however, is unknown. To achieve dynamic RoI location, we propose an dynamic vMFMM clustering algorithm. Our algorithm takes the input of $N \times F \times T$ direction vectors $\mathcal{V}_i$ in time window t_i and output the number of clusters K and vMFMM's parameters Θ. We start with $K = 1$, i.e., one cluster and estimate the vMFMM parameters Θ We define two metrics to measure the quality of the clustering results: minimum inter-cluster distance d which is the smallest angle between two cluster centers and cluster weight w which equals to $min\{a_j\}, 1 \leq j \leq K$. A good clustering result is determined by the below requirement

$$d \geq \mathcal{D} \quad \text{and} \quad w \geq \mathcal{W} \tag{3}$$

in which $\mathcal{D}$ is the threshold to ensure that clusters are as separated as possible and $\mathcal{W}$ is to ensure each cluster has enough amount of data. We increase K by one and carry out another parameter estimation until the condition is not met.

[1] https://youtu.be/8osw3ElPAvY

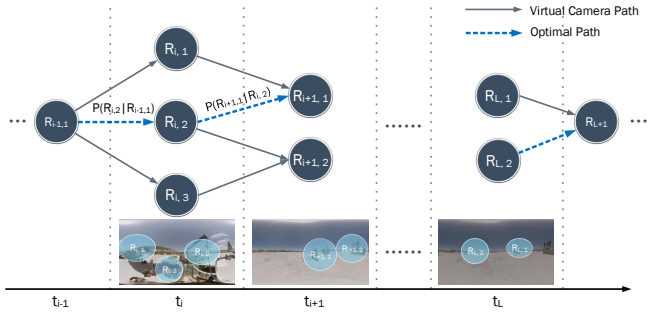

Fig. 3: Bayesian network based chronological order

2.3. Bayesian Network based Attention Transition

In 360 virtual cinematography, users' viewing path or the transition between different RoIs are driven by the progresses of the 360 videos, i.e., the captured events. A high-quality 360 virtual camera paths should follow such logic orders. We propose to use Bayesian network to model the chronological orders between RoIs.

A Bayesian network is a probabilistic graphical model that represents a set of variables and their conditional dependencies via a directed acyclic graph (DAG). In our approach, as shown in Fig. 3, we regard the k_i RoIs $R_{i,j}, 1 \leq j \leq k_i$ located in time window t_i as random variables, and the transition between two RoIs $R_{i,j}$ and $R_{i+1,j'}$ is considered as conditional dependency. The probability of transition between parent RoI $R_{i,j}$ and child RoI $R_{i+1,j'}$ is defined as follows:

$$P(R_{i+1,j'}|R_{i,j}) = \frac{\text{No. of users viewed } R_{i,j} \text{ then } R_{i+1,j'}}{\text{No. of users viewed } R_{i,j}} \tag{4}$$

The transition probabilities between a parent RoI and its children RoI, can then be modeled as a conditional probability table (CPT). Note that in this Bayesian network, two RoIs in the same time window are mutually exclusive, i.e., $P(R_{i,j}, R_{i,j'}|_) = 0, j \neq j'$.

We add a virtual RoI R_0 before the first time window t_1 and R_∞ after the last time window $t_\mathcal{L}$ to represent the start and end of a video. The transition probabilities of the first time window t_1, or the choice of the first RoI is calculated as follows:

$$P(R_{1,j}|R_0) = \frac{\text{No. of users viewed } R_{1,j}}{\text{No. of total users}} \tag{5}$$

And the transition probabilities between RoIs in the last time window and R_∞ is equal to 1.

The joint probability of a camera path with $\mathcal{L}$ RoIs is then defined as follows:

$$P(R_{1:\mathcal{L},_}) = \prod_{i=0}^{\mathcal{L}} P(R_{i+1,j'}|R_{i,j}) \tag{6}$$

Algorithm 1: Optimal Camera Path

Input: $T_{i,j}, 1 \leq i \leq \mathcal{L}, 1 \leq j \leq k_i$
Output: $\{R_0, R_1, \cdots, R_\mathcal{L}\}$
$\mathcal{P}_{\mathcal{L},_} = 1$;
$\mathcal{C}_{\mathcal{L},_} = R_\infty$;
foreach $i \in \{\mathcal{L}, \cdots, 0\}$ **do**
  foreach $j \in \{1, \cdots, k_t\}$ **do**
    Calculate $\mathcal{C}_{i,j}, \mathcal{P}_{i,j}$ according to Eq. 7, Eq. 8;
return `Optimal`$=\{R_0, R_1 = C_{R_0}, R_2 = C_{R_1}, \cdots\}$

2.4. Camera Path Generation

2.4.1. *Dynamic Programming based Path Generation*

We present two ways of determining the virtual camera directions in each time window, `Random` and `Optimal`. In the first method, we randomly choose the next RoI based on the CPT of the current parent RoI and sample a forward vector from the vMF distribution of the selected RoI.

In the second method `Optimal`, we move the virtual camera following the path that has the maximum joint probability, i.e., maximizing Equation 6. We adopt a dynamic programming based algorithm to generate the `Optimal` path, as shown in Algorithm 1.

Algorithm 1 takes input the CPT $T_{i,j}$ of each RoI $R_{i,j}, 1 \leq i \leq \mathcal{L}, 1 \leq j \leq k_i$. The probability of an optimal virtual camera path starting from $R_{i,j}$ is denoted as $\mathcal{P}_{i,j}$, in which the next RoI is denoted as $\mathcal{C}_{i,j}$. Algorithm 1 calculates the `Optimal` virtual camera path in a reverse order, starting from $R_{i,j}, i = \mathcal{L}, 1 \leq j \leq k_i$. The probability and next RoI are calculated using Equation 7 and Equation 8. The algorithm then returns the `Optimal` virtual camera path.

$$\mathcal{C}_{i,j} = \underset{x \in \{1,\cdots,k_{i+1}\}}{\arg\max} \times P(R_{i+1,x}|R_{i,j})\mathcal{P}_{i+1,x} \tag{7}$$

$$\mathcal{P}_{i,j} = P(R_{i+1,\mathcal{C}_{i,j}}|R_{i,j})\mathcal{P}_{i+1,\mathcal{C}_{i,j}} \tag{8}$$

2.4.2. *Kalman Smoother based Camera Stabilization*

After selecting the target RoI in each time window, we then generate the camera position, i.e., the forward vector, f on each video frame. The target RoI's position is used as the forward vector at the beginning of a time window, which is called anchor point. We calculate the intermediate camera positions as a weighted sum of two consecutive anchor points R_i and R_{i+1}, defined as follows:

$$f = (1 - \omega)R_i + \omega R_{i+1} \tag{9}$$

in which ω is the ratio of time lapsed of current window.

The virtual camera paths directly generated from the above algorithm often have jittered movements, causing unpleasant user experience. Therefore we employ a Kalman

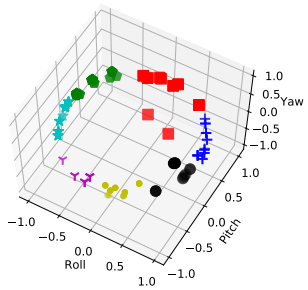

Fig. 4: Dynamic vMFMM

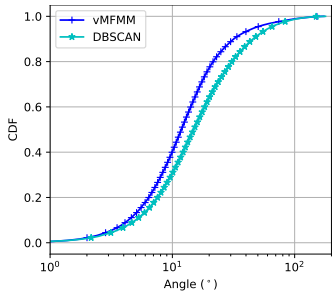

Fig. 5: Angular Distance

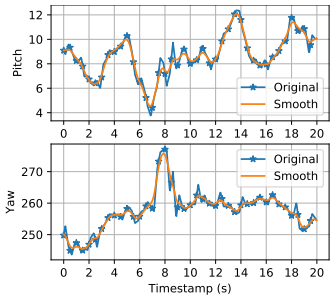

Fig. 6: Kalman Smoother

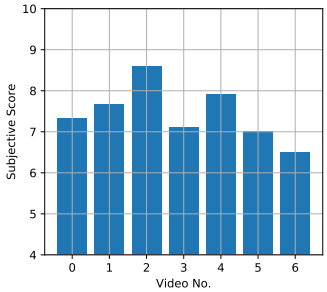

Fig. 7: Subjective Score

smoother[7] to smoothen the movement. A Kalman smoother or filter tries to estimate the real state of a system from inaccurate and uncertain observations. In our approach, we consider the generated camera path as the observations and apply Kalman smoother to obtain the "real" and smoothened paths. A Kalman smoother describes a system with a transition matrix A and an observation matrix H. The transition matrix A defines the state transition relation between past and current state. And the observation matrix H defines the relation between the system state and the observed measurement.

We convert the forward vector to the three-dimensional Euler angle, which represents an object's orientation with rotations around the X, Y and Z axises. We only consider the Yaw rotation α and Pitch rotation β of an euler angle. The two rotations and the corresponding speeds are processed separately by a Kalman smoother to obtain stable movements. After obtaining the horizontal rotation angle α and vertical rotation angle β, we then calculate a rotation matrix following "Yaw first, then Pitch" order. Finally, we obtain the camera direction, i.e., the viewport, on each video frame.

3. EVALUATION

In this section, we first evaluate the algorithms proposed in this paper. We then show the results of virtual cinematography on some of the representative videos.

To demonstrate the effectiveness of our vMFMM based clustering algorithm, we compare it to the conventional dynamic clustering algorithm DBSCAN[2], which is a density-based dynamic clustering algorithm. Specifically, we adopt a spherical DBSCAN method, which uses the angular distance as the distance metric. The DBSCAN algorithm will fine-tune two parameters: $minPts$, the number of minimum samples required to form a group; and ϵ, the maximum density-reachable distance. In our experiments, we set the cluster weight $\mathcal{W}$ of dynamic vMFMM to 0.05 and mimimum samples $minPts$ to $0.05 \times N \times F \times T$. We set both the cluster distance $\mathcal{D}$ of vMFMM and ϵ to $\frac{\pi}{8}$.

In Fig. 4, we show the clustering result of dynamic vMFMM on a highly distributed attention trace. We observe that dynamic vMFMM can achieve a very fine-grained clustering results. We further calculate the distance of a direction vector to its cluster center for the two algorithms. We plot the cumulative distribution function (CDF) plot of the results in Fig. 5. We observe that the clustering results of dynamic vMFMM are more tightly grouped. Compared to DBSCAN, the average angular distance to cluster center is reduced by 21.7%.

Next we demonstrate the effectiveness of Kalman smoother with fine-tuned state noise and observation noise covariance. We show in Fig. 6 the smoothened camera movements compared to the original. We can observe that the original jittered camera movements are perfectly refined to a smooth movement. We further present a demo video in the supplementary material to compare the virtual cinematography results with and without Kalman smoother.

Some demo videos can be downloaded from **Google Drive**[3]. These demos confirm that our method provide accurate and stable cinematography results. Furthermore, we conduct a user study to investigate the subjective evaluation of our method. We invite 12 users to watch the demo videos, and ask them to score each video based on comfort, stableness and narrative. The scores are between 0 to 10, in which 6 represents an acceptable result, as shown in Fig. 7. We can observe that users are satisfied with all the cinematography results. Although for videos with complicated narratives, e.g., 360 sci-fi video, clip 6, the results are slightly lower.

4. CONCLUSION

We propose a data-driven 360 virtual cinematography framework based on user viewing traces. We design a vMFMM based RoI location algorithm and a Bayesian network based camera path generation algorithm. Our method can generate diversified and user-satisfied virtual cinematography results.

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